

The Tile Game

lol why did i make this presentation so formal and stuff

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Background

The Tile Game

- Excursion 1 was the Tile Game.

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The Tile Game

- Excursion 1 was the Tile Game.
- You start with a pile of tiles in 2 colors.
- Players can take any number of tiles of a single color or the same number of tiles from both colors.
- Last person to take a tile wins the game.

Terminology

A position can be represented as a pair $(x, y) \in \mathbb{N}^2$, where x represents the number of tiles of color 1, and y represents the number of tiles of color 2.

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- Winning positions are positions where you can make a move to a losing position.
- Losing positions are positions where all moves you can make go to winning positions.

Terminology

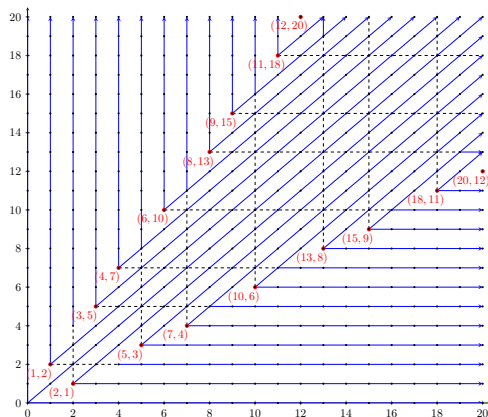
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- Winning positions are positions where you can make a move to a losing position.
- Losing positions are positions where all moves you can make go to winning positions.

We call the set of all losing positions $\mathcal{L}$, and the set of all winning positions $\mathcal{W}$.

That one graph everyone made



Basic Results

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Proposition

There are no positions that are both losing and winning.
That is, $\mathcal{L} \cap \mathcal{W} = \emptyset$. Also, $(0, 0)$ is a losing position.

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Proposition

Every position is either winning or losing. Also, $\mathcal{L}$ and $\mathcal{W}$ form a partition on $\mathbb{N}^2$.

Proposition

If (x, y) is a losing position, then (y, x) is a losing position.

The First Theorem

Theorem

Let $\lambda_n = (x, y)$ be the n th losing position when ordered by $|\lambda| = x + y$, and where $(0, 0)$ is the 0th losing position. Then

$$\lambda_n = (x, x + n).$$

Additionally,

$$x = \min \{a \in \mathbb{N} \mid \forall k < n (\lambda_k = (\alpha, \beta) \implies a \neq \alpha, \beta)\}.$$

More simply, x is the smallest number that has not been “used up” by another losing position, and $y = x + n$.

Post-Rigorous Analysis

Theorem 1 is very important, because it lets us compute losing positions very easily with a Python script.

```
1  #!/usr/bin/env python3
2  avoid = set()
3  losing = []
4  y_int = 1
5  MAX = 1000 # maximum number of tiles on board
6
7  for x in range(1, MAX+1):
8      if x not in avoid:
9          losing.append((x,x + y_int))
10         y_int += 1
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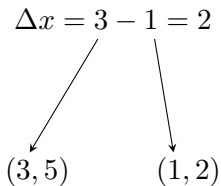
People were looking for patterns in the differences between the number of tiles, but there were no *obvious* patterns. But a pattern exists, and theorem 1 lets us compute enough numbers to find it.

Differences

$(3, 5)$

$(1, 2)$

Differences

$$\Delta x = 3 - 1 = 2$$


$(3, 5)$ $(1, 2)$

Differences

λ_n	Δx	λ_n	Δx	λ_n	Δx
(0, 0)	0	(12, 20)	1	(25, 41)	1
(1, 2)	1	(14, 23)	2	(27, 44)	2
(3, 5)	2	(16, 26)	2	(29, 47)	2
(4, 7)	1	(17, 28)	1	(30, 49)	1
(6, 10)	2	(19, 31)	2	(32, 52)	2
(8, 13)	2	(21, 34)	2	(33, 54)	1
(9, 15)	1	(22, 36)	1		
(11, 18)	2	(24, 39)	2		

Differences

1212212122122122121

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1212212122122122121

- Only 2s and 1s

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121221212212212122121

- Only 2s and 1s
- Never more than one 1 in a row

Differences

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- Only 2s and 1s
- Never more than one 1 in a row
- Never more than two 2s in a row

Differences

121221212212212122121

Differences

121221212212212122121

12122121
 ↓ ↓ ↓
 121

Differences

121221212212212122121

12122121

121



1

Differences

121221212212212122121

12122121

121
↓
1

What if we do the same thing, but in reverse?

Investigation of the Sequence

Formalism

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- We need a formal definition of our sequence.
- “Counting the 2s” is very hard to formalize.
- We need to look for more patterns.
- To do this, we should compute a few of these strings.

Formalism

1

121

12122121

121221212212212122121

Formalism

1
↓
121

12122121

12122121221221221212121

Formalism

1

121


12122121

1212212122122122122121

Formalism

1

121

12122121

121221212212212122121

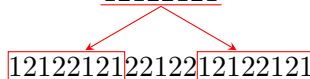


Formalism

1

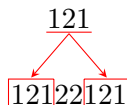
121

12122121



Formalism

1



1212212122122122122121

Formalism



12122121

121221212212212122121

Formalism

Let's call the n th string in this sequence $\Phi(n)$. This means

$$\Phi(1) = 1$$

$$\Phi(2) = 121$$

$$\Phi(3) = 12122121$$

$$\Phi(4) = 12122121221221212121$$

$$\vdots$$

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If our last slide is correct, we should be able to recursively define Φ as

$$\Phi(n) = \Phi(n-1) \oplus \Psi \oplus \Phi(n-1),$$

where Ψ is some unknown string.

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where Ψ is some unknown string.

The $\oplus$ operation is **string concatenation**.

Identifying Ψ

1

121

12122121

121221212212212122121

Identifying Ψ

meeooowoww

2

22

22122

Identifying Ψ

meeooowoww

2
↙ ↘
22

22122

Identifying Ψ

meeooowoww

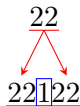
2



Identifying Ψ

meeooowoww

2



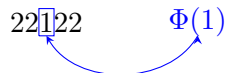
Identifying Ψ

meeooowww

2

22

22122 $\Phi(1)$



Identifying Ψ

$$\Psi(1) = 2$$

$$\Psi(2) = 22$$

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Identifying Ψ

$$\Psi(1) = 2$$

$$\Psi(2) = 22 = \Psi(1) \oplus \Psi(1)$$

$$\Psi(3) = 22122 = \Psi(2) \oplus \Psi(1) \oplus \Psi(2).$$

Identifying Ψ

$$\Psi(1) = 2$$

$$\Psi(2) = 22 = \Psi(1) \oplus \Psi(1)$$

$$\Psi(3) = 22122 = \Psi(2) \oplus \Phi(1) \oplus \Psi(2).$$

What if we try $\Psi(n) = \Psi(n-1) \oplus \Phi(n-2) \oplus \Psi(n-1)$, where $\Phi(0)$ is the empty string and $\Psi(1) = 2$?

Definitions

Definition: (Generating Functions)

Let Φ be the function satisfying

$$\Phi(n) = \begin{cases} 1, & n = 1 \\ \Phi(n-1) \oplus \Psi(n-1) \oplus \Phi(n-1), & n > 1 \end{cases}.$$

Let Ψ be the function satisfying

$$\Psi(n) = \begin{cases} 2, & n = 1 \\ \Psi(n-1) \oplus \Phi(n-2) \oplus \Psi(n-1), & n > 1 \end{cases}.$$

Also let $\Phi(0)$ and $\Psi(0)$ both be the empty string.

Definitions

Definition: (Differences)

Let $\lambda_n = (x_n, y_n)$ and $\lambda_{n-1} = (x_{n-1}, y_{n-1})$. Then define $\Delta x_n = x_n - x_{n-1}$.

Definitions

Definition: (Differences)

Let $\lambda_n = (x_n, y_n)$ and $\lambda_{n-1} = (x_{n-1}, y_{n-1})$. Then define $\Delta x_n = x_n - x_{n-1}$.

Definition: (Losing String)

Let $\mathfrak{X}(n)$ be the string formed from the first n values of Δx . That is,

$$\mathfrak{X}(n) = \bigoplus_{i=1}^n \Delta x_i.$$

The Fundamental Theorem of the Tile Game

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Theorem: (Fundamental Theorem)

For all $n \in \mathbb{N}$, there exists some $\omega \in \mathbb{N}$ such that $\Phi(n) = \mathfrak{X}(\omega)$.

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Theorem: (Fundamental Theorem)

For all $n \in \mathbb{N}$, there exists some $\omega \in \mathbb{N}$ such that $\Phi(n) = \mathfrak{X}(\omega)$.

Corollary

For all $\omega \in \mathbb{N}$, there exists $n \in \mathbb{N}$ such that $\mathfrak{X}(\omega)$ is a substring of $\Phi(m)$ for all $m \geq n$.

Discussion

But does it work?

Before I was able to prove the theorem, I verified it worked up to positions with $400 \cdot 10^6$ tiles in the starting position.

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If you're interested in the proof and the various lemmas required, feel free to ask me about it.

Φ -like Occurances

One interesting thing about the game is the number of places Φ or Φ -like sequences appear.

Fibonacci Connections

One fascinating thing about the problem is its many connections to the Fibonacci numbers.

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Proposition

Let $\|S\|$ be the length of the string S , and let $T(S)$ be the sum of all characters in the string S . Then

$$\|\Phi(n)\| = F_{2n}, \quad \|\Psi(n)\| = F_{2n-1},$$

$$T(\Phi(n)) = F_{2n+1} - 1, \quad T(\Psi(n)) = F_{2n} + 1.$$

Fibonacci Connections

Proposition

For all $n \in \mathbb{Z}^+$,

$$(F_{2n}, F_{2n+1}) \in \mathcal{L},$$

$$(F_{2n+1} - 1, F_{2n+2} - 1) \in \mathcal{L}.$$

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Proposition

Let $(a, b) \in \mathcal{L}$. Then $(a + b, a + 2b) \in \mathcal{L}$.

Fibonacci Connections

Proposition

Let

$$\mathcal{F} = \{(F_{2n}, F_{2n+1}) \mid n \in \mathbb{Z}^+\} \subset \mathcal{L}.$$

Then every losing position $\lambda \in \mathcal{L}$ can be written as

$$\lambda = \sum_{n=1}^N c_n (F_{2n}, F_{2n+1})$$

for some $\{c_n\}_{n=1}^N \subset \mathbb{N}$.

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One remark about proposition 19 is that not all linear combinations of $\mathcal{F}$ are losing positions, but all losing positions are linear combinations of $\mathcal{F}$.

Fibonacci Connections

Corollary

Corollary of proposition 19. Let $(x, y) \in \mathcal{L}$. Then there exists $\{c_n\}_{n=1}^N \subset \mathbb{N}$ such that

$$x + y = \sum_{n=1}^N c_n (F_{2(n+1)}).$$

Fibonacci Connections

Corollary

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Corollary

For all $(x, y) \in \mathcal{L}$ such that $x + y < F_{2n} + F_{2n+1}$, then

$$(x, y) + (F_{2n}, F_{2n+1}) \in \mathcal{L}.$$